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EC : ELECTRONICS AND COMMUNICATIONS
Module 2 : Electronic Devices

INDEX

Sr. No.	Contents	Topics	Pg. No.
1.	**Semiconductor Basics and Diodes**		
	Notes	Introduction	1
		Energy Band Theory	4
		Semiconductor Materials	7
		Carrier Concentration and Mathematical Derivation	11
		Mass–Action Laws	16
		Drift and Diffusion Currents	17
		Carrier Generation and Recombination	20
		Hall Effect	21
		P–N Junction as a Diode	23
		VI Characteristics of Diode	26
		Junction Temperature Effect	27
		Breakdown Voltage	27
		Resistance Levels	28
		Capacitance in P–N Junction	30
		Diode Switching Times	32
		Breakdown Diodes or Zener Diodes	32
		List of Formulae	34
		Important Terms	35
		LMR (Last Minute Revision)	36
	Assignment – 1	Questions	37
	Test Paper – 1	Questions	40

Sr. No.	Contents	Topics	Pg. No.
2.	**Bipolar Junction Transistor**		
	Notes	Introduction	**43**
		Transistor Structure	**43**
		Region of Operation	**44**
		Transistor Currents	**44**
		Transistor Configuration	**48**
		Non-Linear Transistor Leakage Currents and Breakdown Voltage	**55**
		Transistor as a Switch	**58**
		Transistor Switching Characteristics	**58**
		List of Formulae	**59**
		Important Symbols and Terms	**60**
		LMR(Last Minute Revision)	**60**
	Assignment – 2	Questions	**61**
	Test Paper – 2	Questions	**65**
3.	**Field Effect Transistor**		
	Notes	Introduction	**69**
		Classification	**69**
		Metal Oxide Semiconductor Field Effect Transistor (MOSFET)	**70**
		Enhancement MOSFET(E - MOSFET)	**70**
		Depletion MOSFET (D–MOSFET)	**72**
		Difference Between JFET and MOSFET	**74**
		The ideal MOS–Capacitor	**74**
		List of Formulae	**76**
		Important Notations	**76**
		LMR (Last Minute Revision)	**77**
	Assignment – 3	Questions	**78**
	Test Paper – 3	Questions	**81**

Sr. No.	Contents	Topics	Pg. No.
4. Miscellaneous Devices			
	Notes	Light Emitting Diodes (LED)	**84**
		LMR(Last Minute Revision)	**86**
Practice Problems		Questions	**87**
ID Problems		Questions	**97**
SOLUTIONS			
Assignments		Answer Key	**102**
		Model Solutions	**103**
Test Papers		Answer Key	**114**
		Model Solutions	**115**
Practice Problems		Answer Key	**126**
		Model Solutions	**127**
ID Problems		Answer Key	**138**
		Model Solutions	**139**

Topic 1 : Semiconductor Basic and Diodes

INTRODUCTION

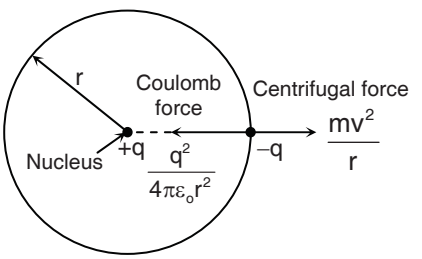
An electronic device controls the movement of electrons. The study of electronic devices requires a basic understanding of the relationship between electrons and other components of an atom. This leads to a knowledge of differences between conductors, insulators and semi conductors.

- **Atomic Theory**

  In order to under stand energy band structure, we first study basic atomic theory and electronic configuration. In the year 1911, Rutherford established that an atom consists of positively charged nucleus and a number of negatively charged electrons which revolve round the nucleus in various orbits. The nucleus may be considered to be built up of a number of neutral particles (neutrons) and a number of positively charged particles, called protons. In each atom the number of revolving electrons is equal to z, the number of protons in the nucleus. Hence atom is electrically neutral.

Hydrogen Atom

A hydrogen atom consists of a single proton at the nucleus and an electron moving around the nucleus. The force of attraction between the positive nucleus and the negative electron follows the Coulomb's law. The resultant path followed by the electron, in accordance with the classical mechanics, is then either a circle or an ellipse. For the sake of simplicity, let us assume that this electron revolves as a point-like article in a circular orbit of radius *r* with nucleus at the centre. If the electron is stationary, it would fall into the nucleus. Hence it is necessary that this electron must revolve around the nucleus and have acceleration towards the nucleus.



Circular orbit of an electron in a hydrogen atom

Such that centrifugal force = Coulomb force

$$\frac{mv^2}{r} = \frac{q^2}{4\pi\varepsilon_0 r^2} \qquad \ldots(1)$$

where v is the velocity of the revolving electron and ε_0 is the absolute dielectric constant (8.854×10^{-12} Farad/meter).

Thus Revolving Electron Possesses :

- Kinectic energy = $\frac{1}{2}mv^2$
- Potential energy $U = \frac{-q^2}{4\pi\varepsilon_0 r}$ Joules
- The total energy of the electron is then given by,

$$W = \frac{1}{2}mv^2 - \frac{q^2}{4\pi\varepsilon_0 r} \qquad \ldots(2)$$

Combining equations (1) and (2), we get

$$W = -\frac{q^2}{8\pi\varepsilon_0 r} \qquad \ldots(3)$$

The minus sign indicates that the electron is bound to the nucleus and it requires amount of energy equal to $-W\left(\text{or}\,\frac{q^2}{8\pi\varepsilon_0 r}\right)$ to remove the electron from the nucleus.

- **The Bohr Atom**

  The above discussion falls completely within the domain of classical physics. However, this classical atomic model is unstable since the energy of the electron decreases continually as a result of emission of electromagnetic radiation, with the end result that the electron would spiral into the nucleus. In order to avoid this pitfall and to retain the stability of the orbit, Bohr postulated the following three fundamental laws :

  1. All energy levels as given by classical mechanics are not possible; the electron can possess only certain discrete energies. When an electron occupies a state corresponding to one of these discrete energies, the electron does not emit radiation and is said to be in *stationary* of *non-radiating* state.
  2. Transition of an electron from one stationary state corresponding to energy W_2 to another stationary state with energy W_1 results in emission of radiation. The frequency of this radiant energy is given by,

     $$f = \frac{W_2 - W_1}{h} \qquad \ldots(4)$$

     where f is the frequency in cycles/second or hertz, h is the Planck's constant in joule-second and equal 6.62×10^{-34} joule-second, and W_1 and W_2 are expressed in joules.
  3. A stationary or non-radiating state must meet the condition that the angular momentum of the electron in this state is quantised and must equal an integral multiple of $h/2\pi$

     Thus $$mvr = \frac{nh}{2\pi} \qquad \ldots(5)$$

     there n is an integer.

     Combining equations (1) and (5), we get the values of radii of stationary states.

     Thus, radius $$r_n = \frac{\varepsilon_0 h^2}{\pi . mq^2} n^2 \qquad \ldots(6)$$

     $$= 0.529 \times 10^{-10}\ n^2 \text{ metre}$$

     Combining equation (3) and (6), and on substituting the values of m, q, ε_0 and h we get, $$\boxed{E_n = -\frac{13.6}{n^2} \text{ electron volts}} \qquad \ldots(7)$$

     From equation (7) we conclude that in its lowest state (n = 1), called the *ground state* or the *normal state*, the electron is bound to the proton by energy of 13.6 eV. Thus it requires 13.6 eV to ionize the hydrogen atom.

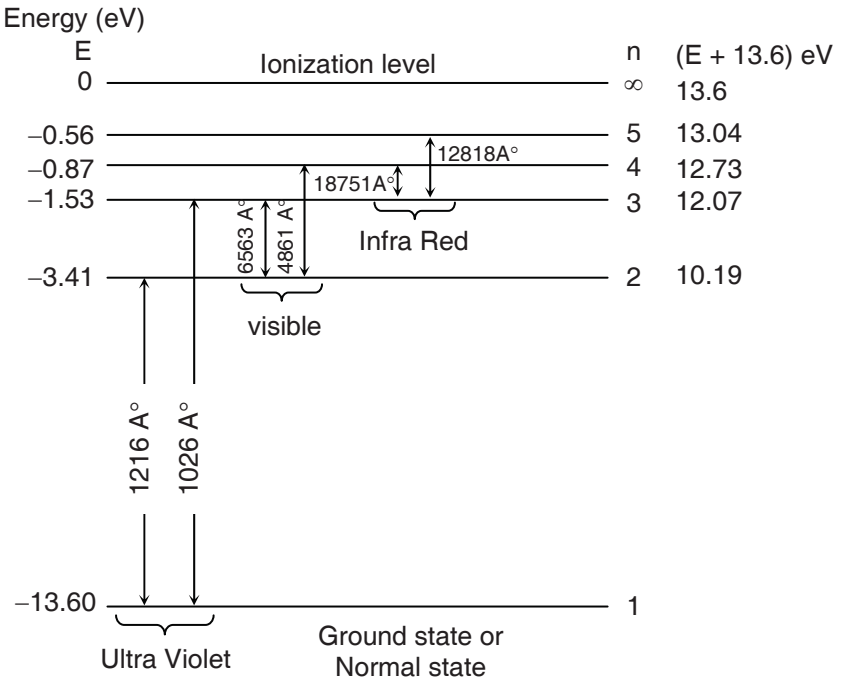
     Knowing the energy levels E_1, E_2 etc. of stationary states as given by equation (7), we may be making use of equation (4) to find the exact frequencies in the hydrogen spectrum.

     The value of energy of the stationary states is customarily expressed in electron volts E, rather than in joules W. Also more commonly, the emitted radiation is specified by its wavelength λ in angstroms rather than by its frequency f in hertz. Making use of these units, equation (4) may be rewritten as below,

     $$\lambda = \frac{12,400}{(E_2 - E_1)}$$ where E_2 and E_1 are the energy levels in electron volts.

Atomic Energy Level Diagram

Following figure shows, the so-called, energy level diagram for hydrogen. When the energy is large enough, the electron may move completely out of the field of the nucleus and thus get detached from the nucleus. The minimum energy required for this process is called the *ionization energy* and this represents the highest state in the energy level diagram. Thus for hydrogen, ionization energy equals 13.6 eV.



Energy level diagram of hydrogen

Electronic Structure of Elements

Electronic configuration can be uniquely specified using four quantum number as

(i) the principle quantum number n = 1, 2, 3, …

(ii) the angular momentum quantum number ℓ = 0, 1, 2, …, (n – 1)

(iii) the magnetic quantum number m_l = 0, ±1, ±2, …, $\pm\ell$.

(iv) the spin quantum number $m_s = \pm 1/2$.

The principle quantum number n plays the same role as the quantum number n in the atomic theory of Bohr, i.e. it gives the energy level. The angular momentum quantum number ℓ determines the angular momentum of the electron while the magnetic quantum number m_ℓ determines the component of the angular momentum along the prescribed direction which may be the direction of an external magnetic field.

- **Pauli Exclusion Principle**

  This law was enunciated by pauli in the year 1925. *Pauli exclusion principle states* that no two electrons in an electronic system can have the same set of four quantum numbers n, ℓ, m_ℓ and m_s, i.e. no two electrons can occupy the same quantum state.

  The periodic table of the chemical elements may be explained by pauli exclusion principle. Depending on this, electronic configuration of every atom has different

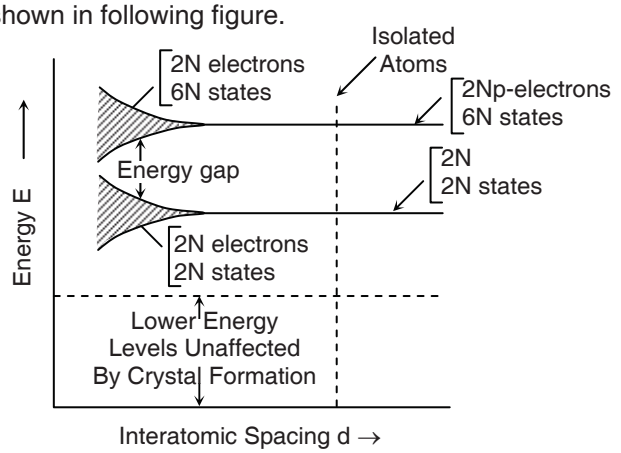
shells and subshells and the number of electrons in each shells are given in the following table.

- **Electron Shells and Sub-Shells**

Shell	K	L		M			N			
n	1	2		3			4			
Sub-shell ℓ	s	s	p	s	p	d	s	p	d	f
	0	0	1	0	1	2	0	1	2	3
m_ℓ	0	0	0, ±1	0	0, ±1	0, ±1, ±2	0	0, ±1	0, ±1, ±2	0, ±1, ±2, ±3
Number of States	2	2	6	2	6	10	2	6	10	14
	2	8		18			32			

ENERGY BAND THEORY

It has been found that most of the metals and semiconductors have crystalline structure. In the case of a gas, the atoms are for apart so as to have negligible influence on each other. Hence the electronic energy levels for single free atoms apply in the case of gas also. However, in the case of a crystal, the atoms are closely packed. Thus the energy band levels for single atoms do not apply because the potential is now a periodic function in space, its value at any point in space resulting from the contribution by every atom. Because of these the energy levels of the outer orbit electrons are altered sufficiently, which result in coupling between the outer shell electrons, the energy states spread up into a *band* as shown in following figure.

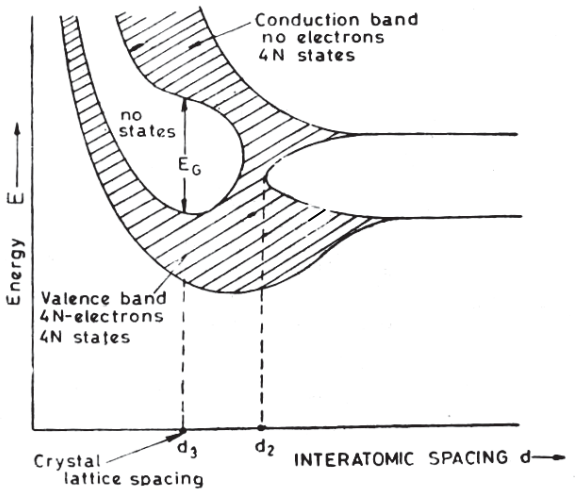


Splitting of energy levels of isolated atoms to form energy band when interatomic spacing is reduced

To study the formation of energy band qualitatively, let us consider a crystal consisting of N atoms say germanium.

- For large interatomic spacing such as d_1 in above figure, the interaction between the atoms is negligibly small, so that the energy levels remain the same as those of an isolated atom.
- For germanium, the outermost two sub-shells contain two s-electrons and two p-electrons. For large inter atomic spacing such as d_1, there are 2N electrons completely filling the 2N possible s-states at the same energy level. In contrast, the p-atomic sub-shell has 6 possible states. Hence out of 6N possible p-states, only 2N states are occupied by 2N electrons, all at the same energy level.

- Next let us decrease the interatomic spacing d in the imaginary crystal. Then each atom will exert an electric force on its neighbours. As a result of this coupling between atoms, the wave functions of the atoms overlap.
- The net effect of the coupling is that 2N s-states spread out to occupy a range of energy bands. Now since N is very large, of the order of $10^{23}/cm^3$, the total energy spread between the maximum and minimum energy levels may become large, several electron volts for small inter-atomic spacing. These numerous discrete but very closely spaced energy levels constitute the so-called *energy band*.
- This energy band is shown schematically as the lower shaded region in above figure. Obviously all the 2N states in this band are completely filled by 2N electrons.
- Similarly at a low value of interatomic spacing, at p–level, 6N, states spread to form a band as shown by the upper shaded region in above figure. However, in this band, out of 6N states, only 2N states are occupied by 2N electrons, leaving 4N states unoccupied.
- Between the two bands shown in above figure, there exists an energy gap constituting the *forbidden band*. No electrons can occupy states in this forbidden band.



Energy band diagram for a crystal

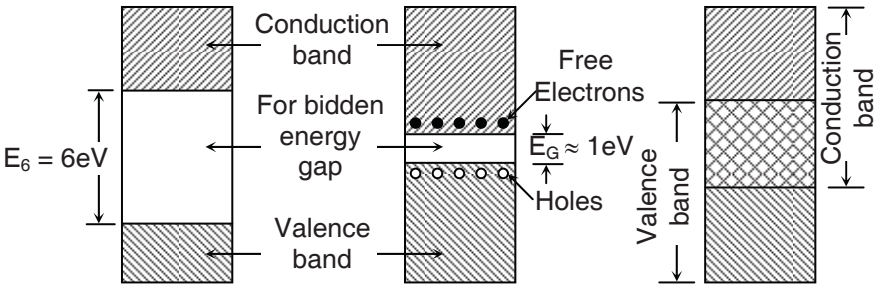
As this interatomic spacing d is progressively decreased, the forbidden band becomes zero for spacing such as d_2 in above figure. For still lower values of d, the two energy bands overlap as shown in above figure.

- In this condition, the 6N states in the upper band merge with the 2N states of the lower band.
- The total number of states in this combined band equals 8N, out of which only half, i.e. 4N states are occupied by available 4N electrons.
- For this interatomic spacing d_2, each atom has contributed 4 electrons to this new combined band. Then 4 electrons now no longer belong to either the s-subshell or the p-subshell; rather they below to the combined new band.
- The element has thus contributed 4 electrons / atom to the crystal in the highest occupied band, called the *valence band* and is, therefore, tetravalent.
- For the crystal lattice spacing d_3, above figure shows that the valence band is completely filled with 4N electrons and this band is separated from the above unfilled

or empty band by a forbidden band which contains no allowed states. This upper band contains no electrons but has 4N states. This upper band is thus empty and is referred to as the *conduction band*.

Energy Band Structure of Insulators, Semiconductors and Metals

- **Insulators** : Following figure (a) gives the energy band structure of a typical insulator.



(a) Insulator (b) Semiconductor (c) Metal

Energy band structure of an insulator, a semiconductor and a metal

For diamond, the forbidden energy gap E_G is rather large, about 6 electron volts as shown in above figure (a). For this high value of E_G it is normally not possible to transfer an electron from valence band to the conduction band. Electrical conduction is thus impossible. Hence diamond constitutes an insulator.

- **Semiconductors**

A semiconductor has a small forbidden energy gap of about 1 eV. The two most important semiconductors used in semiconductor devices are germanium and silicon having E_G of 0.785 and 1.21 eV respectively at 0° Kelvin. Additional energies equal to these values of E_G even cannot ordinarily be obtained from applied fields. Hence these materials in pure or intrinsic form behave as insulators at low temperatures. However, the conductivity increases with the increase of temperature, since E_G decreases as shown by the following formulae.

For Germanium, $E_G(T) = 0.785 - 2.23 \times 10^{-4} T$ …(8)

For silicon, $E_G(T) = 1.21 - 3.60 \times 10^{-4} T$ …(9)

As per equations (8) and (9), the value of E_G at room temperature of 300°K becomes 0.72 eV and 1.1 eV for germanium and silicon respectively.

Since only a few such electrons have shifted to the conduction band, the conductivity of the material is small and the material is referred to as a semiconductor. The above discussion refers to a pure or intrinsic semi-conductor. A pure semiconductor has poor conductivity and is not of much use in formation of semi-conductor devices. The conductivity of a semi-conductor may be greatly increased by adding a very small amount of impurity to the otherwise pure semi-conductor.

The process is known as *doping* and consists in adding impurity atoms of either trivalent or pentavalent material to pure silicon or germanium. As a result, allowable energy states appear in the forbidden energy gap. These impurity levels also contribute to the conduction. Such semi-conductor materials with adequate doping are referred to as *impurity semi-conductors* or *extrinsic semi-conductors*.

- **Metals**
  A metal contains a partly filled band structure. On application of an electric field, the electrons may acquire additional energy and move into higher energy states. Thus mobility of electrons is possible. Since mobile electrons constitute an electric current, this material forms a good conductor. The partly filled regions form the conduction band. Above figure (c) shows the band structure of a metal with overlapping valence and conduction bands.

	Gap energies	
	0°K	**300°K**
Silicon	1.21 eV	1.l eV
Germanium	0.785 eV	0.72 eV

SEMICONDUCTOR MATERIALS

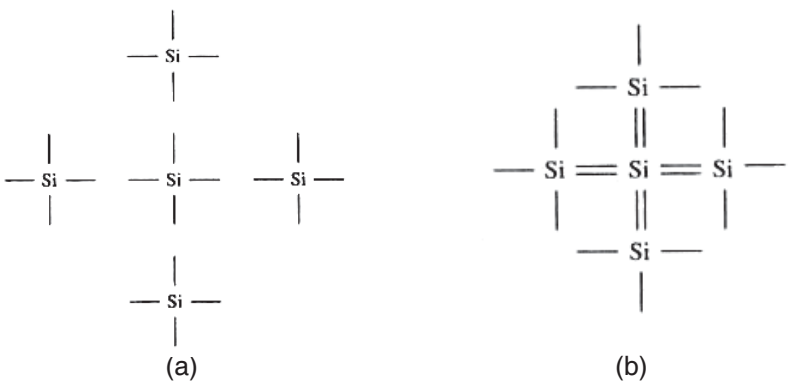
Most electronic devices are fabricated by using semiconductor materials along with conductors and insulators. To gain a better understanding of the behavior of the electronic devices in circuits, we must first understand a few of the characteristics of the semiconductor material.

(A) Intrinsic Semiconductors

An *intrinsic semiconductor* is a single-crystal semiconductor material with no other types of atoms within the crystal.

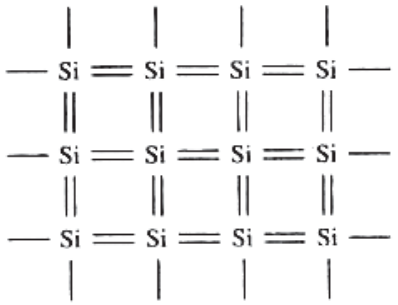
Following figure (a) shows five non-interacting silicon atoms, with the four valence electrons of each atom shown as dashed lines originating from the atom. As silicon atoms come into close proximity to each other, the valence electrons interact to form a crystal. The valence electrons are now shared between atoms, forming what are called *covalent bonds*.

Following figure (b) is a two-dimensional representation of the lattice formed by the five silicon atoms shown in figure (a)



Silicon atoms in a crystal matrix :
(a) five non–interacting silicon atoms, each with four valence electrons
(b) a two–dimensional representation showing the covalent bonding.

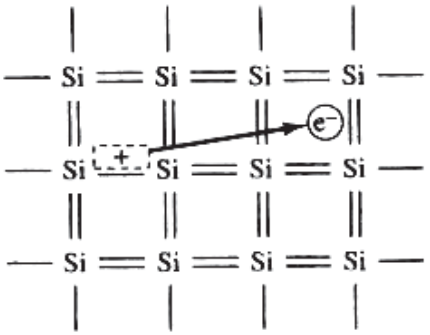
- A two-dimensional representation of a silicon single crystal is shown in figure 1, for T = 0°K, where T = temperature.
- Each line between atoms represents a valence e1ectron and at T = 0°K, each electron is in its lowest possible energy state, so each covalent bonding position is filled.
- If a small electric field is applied to this material, the electrons will not move, because they will still be bound to their individual atoms.
- Therefore, at T = 0°K, silicon is an *insulator*; that is, no charge flows through it.



Two-dimensional representation of the silicon crystal at T = 0 °K

Fig. 1

If the temperature increases, the valence electrons will gain thermal energy. Any such electron may gain enough thermal energy to break the covalent bond and move away from its original position (Figure 2). The electron will then be free to move within the crystal.

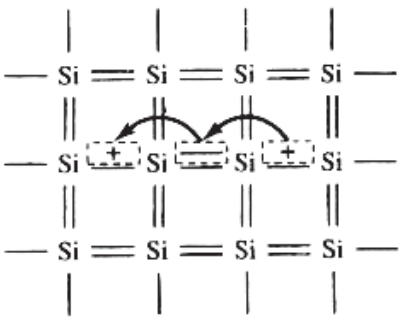


The breaking of a covalent bond for T > 0°K

Fig. 2

Since the net charge of the material is neutral, if a negatively charged electron breaks its covalent bond and moves away from its original position, a positively charged "empty state" is created at that position (Figure 2). As the temperature increases, more covalent bonds are broken and more free electrons and positive empty states are created.

The net flow of these free electrons in a semiconductor causes a current. In addition, a valence electron that has a certain thermal energy and is adjacent to an empty state may move into that position; as shown in figure 3 making it appear as if a positive charge is moving through the semiconductor. This positively charged "particle" is called a *hole*. In semiconductors, then, two types of charged particles contribute to the current: the negatively charged free electron, and the positively charged hole.



A two-dimensional representation of the silicon crystal showing the movement of the positively charged hole

Fig. 3

Intrinsic Carrier Concentration

The concentrations (#/cm)^3 of electrons and holes are important parameters in the characteristics of a semiconductor material, because they directly influence the magnitude of the current. In an intrinsic semiconductor, the densities of electrons and holes are equal, since the thermally generated electrons and holes are the only source of such particles. Therefore, we use the notation n_i as the *intrinsic carrier concentration* for the concentration of the free electron, as well as that of the holes. The equation for n_i is as follows :

$$n_i = BT^{3/2} e^{\left(\frac{-E_g}{2kT}\right)}$$

where B is a constant related to the specific semiconductor material, E_g is the band gap energy (eV), T is the temperature (°K), and k is Boltzmann's constant (86×10^{-6} eV/°K)

Example:

Calculate the intrinsic carrier concentration in silicon at T = 300 °K.

Solution :

For silicon at T = 300°K, we can write

$$n_i = BT^{3/2} e^{\left(\frac{-E_g}{2kT}\right)} \quad \ldots [B = 5.23 \times 10^{15}, Eg = 1.1 \text{ eV}]$$

$$= (5.23 \times 10^{15})(300)^{3/2} e^{\left(\frac{-1.1}{2(86\times10^{-6})(300)}\right)}$$

or $n_i = 1.5 \times 10^{10}$ cm^{-3}.

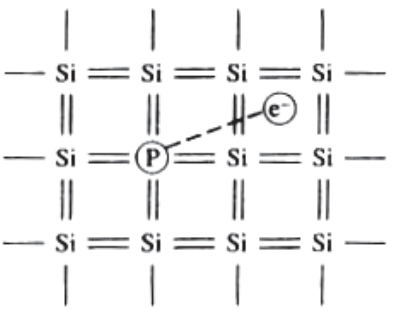
Note : Though this concentration may appear large, but is relatively small compared to concentration of Si atoms which is 5×10^{22} cm^{-3}

(B) Extrinsic Semiconductors

Because the electron and hole concentrations in an intrinsic semiconductor are relatively small, only very small current are possible. However, these concentration can be greatly increased by adding controlled amounts of certain impurities. A desirable impurity is one that enters the crystal lattice and replaces (i.e., substitutes for) one of the semiconductor atoms, even though the impurity atom does not have the same valence electron structure.

(i) *n–Type Semiconductor* :

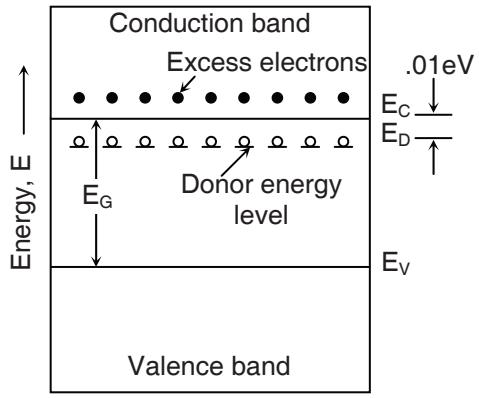
The most common group V elements used for this purpose are phosphorus and arsenic. For example, when a phosphorus atom substitutes for a silicon atom, as shown in figure, four of its valence electrons are used to satisfy the covalent bond requirements. The fifth valence electron is more loosely bound to the phosphorus atom. At room temperature, this electron has enough thermal energy to break the bond, thus being free to move through the crystal and contribute to the electron current in the semiconductor.



Two-dimensional representation of a silicon lattice doped with a phosphorus atom

The phosphorus atom is called a *donor impurity*, since it denotes an electron that is free to move. Although the remaining phosphorus atom has a net positive charge, the atom is immobile in the crystal and cannot contribute to the current. Therefore, when a donor impurity is added to a semiconductor, free electrons are created without generating holes. This process is called *doping*, and it allows us to control the concentration of free electrons in a semiconductor. A semiconductor that contains donor impurity atoms is called an *n-type semiconductor* (for the negatively charged electrons)

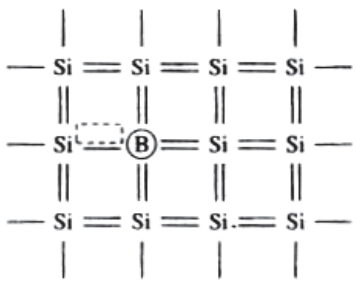
Thus addition of donor impurities to a semiconductor results in introduction of allowable energy levels slightly below the conduction band as shown in following figure. In Ge (Si), this new donor level is about 0.01 eV (0.05 eV) below the conduction level. Hence at room temperature, almost all the excess electrons donated by the donor atoms get raised into the conduction band.



Energy band diagram of n-type semiconductor

(ii) *p–Type Semiconductor* :

The most common group III element used for silicon doping is boron. When a boron atom replaces a silicon atom, its three valence electrons are used to satisfy the covalent bond requirements for three of the four nearest silicon atoms (shown in the adjoining figure). This leaves one bond position open. At room temperature, adjacent silicon valence electrons have sufficient thermal energy to move into this position, thereby creating a hole. The boron atom then has a net negative charge, but cannot move, and a hole is created that can contribute to a hole current.



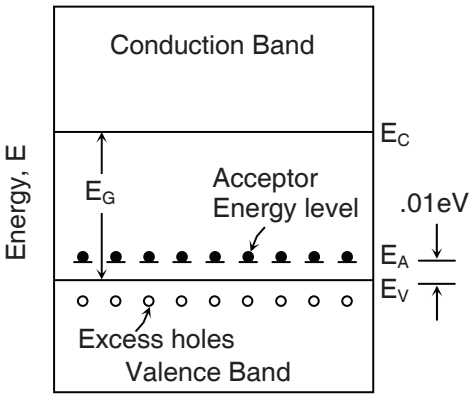
Two-dimensional representation of a silicon lattice doped with a boron atom

Because the boron atom has accepted a valence electron, the boron is therefore called an *acceptor impurity*. Acceptor atoms lead to the creation of holes without electrons being generated. This process, also called doping; can be used to control the concentration of holes in a semiconductor.

A semiconductor that contains acceptor impurity atoms is called a *p-type semiconductor* (for the positively charged holes created).

With the addition of p–type impurity, there get created allowable discrete energy levels just above the valence band as shown in figure. An extremely small amount of energy (0.01 eV) is required to move an electron from the valence band to the acceptor level.

Hence all the acceptor atoms ionize resulting in occupation of all acceptor levels by electrons and creation of holes in the valence band. These holes then constitute the largest number of carriers in the semi-conductor material.



Energy band diagram of p-type semi-conductor

The materials containing impurity atoms are called *extrinsic semiconductor* or *doped semiconductors***.** The doping process which allows us to control the concentration of free electrons and holes, determines the conductivity and currents in the material.

CARRIER CONCENTRATION AND MATHEMATICAL DERIVATION

Energy Distribution

The number of free electrons per cubic metre whose energies lie in the energy interval dE is given by,

$$dn_E = \rho_E \, dE \qquad \ldots(10)$$

where ρ_E represents the density of electrons in this interval.

Function ρ_E represents the distribution in energy and is expressed as the number of electrons per eV per cubic metre of metal.

This function ρ_E may be expressed as,

$$\rho_E = f(E)\, N(E) \qquad \ldots(11)$$

where N(E) is the density of states in the conduction band and is expressed as number of states per eV per cubic metre, and f(E) is the probability that a quantum state with energy E is occupied by an electron.

N(E) is given by,

$$N(E) = \gamma E^{1/2} \qquad \ldots(12)$$

where γ is a constant defined as below,

$$\gamma \equiv \frac{4\pi}{h^3}(2m)^{1/2}(1.60\times10^{-19})^{3/2} = 6.82\times10^{27} \qquad \ldots(13)$$

where dimensions of γ are $(m^{-3})\,(eV)^{-3/2}$, m is the mass of an electron in kilogram and h is Planck's constant in Joule-sec.

Fermi-Dirac Function

The expression for f(E) is called the *Fermi-Dirac probability function* and given the fraction of all possible states at energy E(eV) occupied under conditions of thermal equilibrium. From quantum statistics, the following expression for f(E) is obtained,

$$f(E) = \frac{1}{1+\varepsilon^{(E-E_F)/kT}} \qquad \ldots(14)$$

where T is the temperature in deg. K, k is the Boltzmann constant, eV/deg. K and E_F is the *Fermi level* or *characteristic energy* for the crystal in eV.

- **Carrier Concentrations in an Intrinsic Semiconductor**

  We are required to know the concentration of free electrons n and the concentration of holes p in a semiconductor in order that we may calculate the conductivity.

  (i) *Number of Electrons in the conduction Band*

  Combining equation (10) and (11), we get the following expression for the number of conduction electrons per cubic metre having energies between E and (E + dE),

  $$dn = N(E)\,f(E)\,dE$$

  where E is in electron volts.

  Equation (12) giving the expression for the density of states N(E) is based on the assumption that the bottom of the conduction band is at zero potential. This has to be modified for a semiconductor. In a semiconductor, the lowest energy in the conduction band in E_C. Accordingly equation (12) is modified as below for a semiconductor,

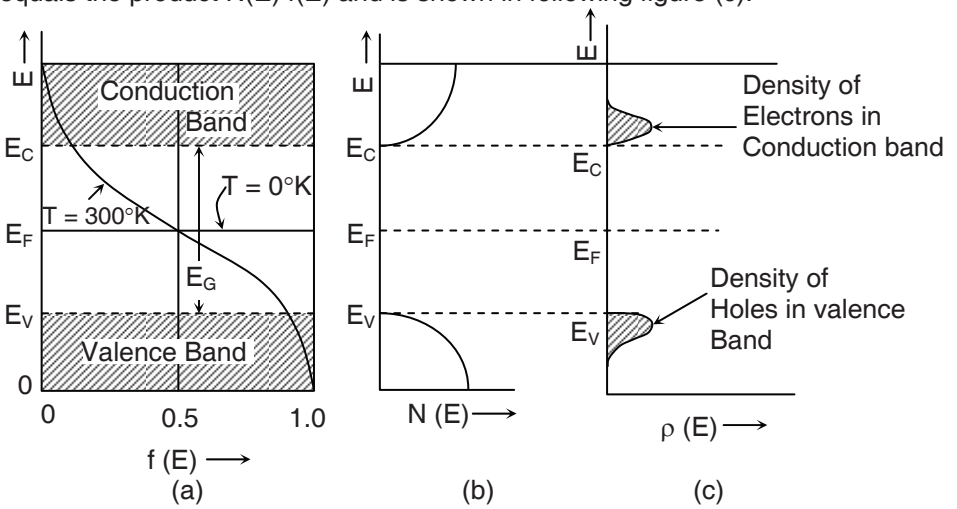
  $$N(E) = \gamma(E - E_C)^{1/2} \text{ for } E > E_C \qquad \ldots(15)$$

  The Fermi function f(E) is given by equation (14) reproduced below,

  $$f(E) = \frac{1}{1+\varepsilon^{(E-E_F)/kT}} \qquad \ldots(16)$$

  The figure below gives the energy band diagram of an intrinsic semiconductor with the Fermi-Dirac probability function f(E) for 0°K and 300°K superimposed on it. Evidently at 0°K, the probability of finding an electron in the conduction band is zero i.e. f(E) = 0, while the probability of finding a hole in the valence band is zero i.e. 1 – f(E) = 0. At room temperature (300°K), on the other hand, a few electrons get excited to higher energies so that a few states near the bottom of the conduction band get filled as shown by the curve for 300°K in following figure (a). Further at room temperature, in the upper region of the valence band, the probability of occupancy of states gets decreased slightly below unity since some of the electron have escaped from the covalent bond and have shifted to the conduction band. Following figure (b) shows the curve giving the density of states

N(E) in conduction band as given by equation (15). The density of electrons equals the product N(E) f(E) and is shown in following figure (c).



(a) Fermi Dirac Probability function f(E) at 0°K and 300°K superimposed on the energy band diagram for intrinsic semiconductor.
(b) Density of states N(E) in conduction and valence bands.
(c) Density of carriers ρ(E) = N(E) f(E) for electrons and N(E) [1–(f(E)] for holes at room temperature (300°K)

Density of states and density of carriers in intrinsic semiconductor

The concentration of electrons in the conduction band is nothing but the area under this curve of above figure (c) and is given by,

$$n = \int_{E_C}^{\infty} N(E)\, f(E)\, dE \qquad \ldots(17)$$

For $E \geq E_C$, i.e., in the conduction band, $E - E_F \gg kT$. Hence equation (16) yields,

$$f(E) = \varepsilon^{-(E-E_F)/kT} \qquad \ldots(18)$$

Combining equations (17), (18) and (15), we get

$$n = \int_{E_C}^{\infty} \gamma (E - E_C)^{1/2}\, \varepsilon^{-(E-E_F)/kT} dE \qquad \ldots(19)$$

On substituting the value of γ and integrating, we get

$$n = N_C\, \varepsilon^{-(E_C-E_F)/kT} \qquad \ldots(20)$$

$$\text{where } N_C = 2\left(\frac{2\pi m_n kT}{h^2}\right)^{3/2} (1.60 \times 10^{-19})^{3/2} = 2\left(\frac{2\pi m_n \bar{k}T}{h^2}\right)^{3/2} \qquad \ldots(21)$$

where k is in electron volts per degree kelvin while $\bar{k}$ is in joules per degree kelvin. Further m_n replaces m and represents the *effective mass* of the electron.

(ii) *Number of Hole in the Valence Band*

On similar lines, we can find number of holes in valence bands and is given as

$$p = Nv\,\varepsilon^{-(E_F - E_V)/kT} \qquad \ldots(22)$$

where $$N_V = 2\left(\frac{2\pi m_p kT}{h^2}\right)^{3/2} (1.60 \times 10^{-19})^{3/2}$$

$$= 2\left(\frac{2\pi m_p \bar{k}T}{h^2}\right)^{3/2} \qquad \ldots(23)$$

Equations (23) giving N_V is the same as equation (21) giving N_C except that m_n has been replaced by m_p, the effective mass of a hole.

- **Fermi Level in an Intrinsic Semiconductor**

Equations (20) and (22) giving respectively the density of electrons in conduction band and the density of holes in valence band apply equally to intrinsic as well as extrinsic semiconductors. To distinguish between the two categories of semiconductors, we add subscript i to n and p. Thus n_i and p_i pertain to intrinsic semiconductor. Now since the semiconductor crystal is electrically neutral,

$$n_i = p_i$$

Hence equations (20) and (22) yield,

$$N_C\ \varepsilon^{-(E_C - E_F)/kT} = N_V\ \varepsilon^{-(E_F - E_V)/kT}$$

On taking logarithms of both sides, we get

$$\ln\frac{N_C}{N_V} = \frac{E_C + E_V - 2E_F}{kT}$$

Hence $$E_F = \frac{E_C + E_V}{2} - \frac{kT}{2}\ln\frac{N_C}{N_V} \qquad \ldots(24)$$

If the effective mass of electron equals the effective mass of hole i.e., if $m_n = m_p$, then $N_C = N_V$ and hence equation (24) gives,

$$E_F = \frac{E_C + E_V}{2} \qquad \ldots(25)$$

Equation (25) shows that the Fermi level E_F lies in the centre of forbidden energy gap, as depicted in above figure (see the page 11).

Fermi Level as a Function of Temperature

(i) *n-Type Semiconductor*

In an n-type semi-conductor, density of donor atoms N_D exceeds the density of acceptor atoms N_A.

$\therefore \quad n \approx N_D$

Then from equation (20), we get

$$N_D = N_C\ \varepsilon^{-(E_C - E_F)/kT}$$

Hence $$E_F = E_C - kT\ \ln\frac{N_C}{N_D} \qquad \ldots(26)$$

Equation (26) shows that the Fermi level E_F lies below E_C by a value depending on the temperature T and density N_D of donor atoms.

(ii) *p-Type Semiconductor*

In p-type semi-conductor, the following assumptions are made in the derivation of expression for Fermi level :

- The density of acceptor atoms far exceeds the density of donor atoms i.e. $N_A \gg N_D$.
- Every acceptor atom has accepted one electron from the valence band.
- The density of electrons in the conduction band is much smaller then the density of holes in the valence band.

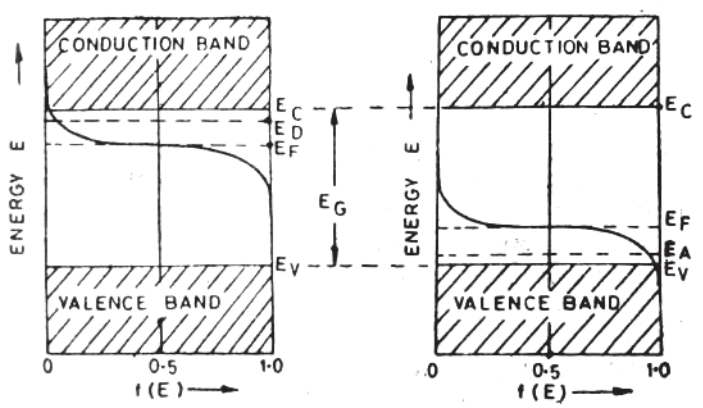
$\therefore \quad p \approx N_A$

Then from equation (22), we get

$$N_A = N_V \, \varepsilon^{-(E_F - E_V)/kT}$$

or $$E_F = E_V + kT \ln \frac{N_V}{N_A} \qquad \ldots(27)$$

Equation (27) shows that the Fermi level E_F lies above the top of the valence band by an amount depending upon the temperature T and concentration of acceptor atoms N_A. Following figure shows the shifted positions of Fermi level in n-type and p-type semi-conductors



(a) n-type (b) p-type

Shifted positions of Fermi level in n-type and p-type semi-conductors

Example

In an n-type semiconductor, the Fermi level lies 0.3 eV below the *conduction band at* 300°K. If the temperature is *increased* to 330°K, find the new position of Fermi level.

Solution :

$$E_F = E_C - kT \ln \frac{N_C}{N_D}$$

Hence $$(E_C - E_F) = kT \ln \frac{N_C}{N_D}$$

Hence $$0.3\,\text{eV} = 300\,k \cdot \ln \frac{N_C}{N_D}$$

For $T = 330°K$, $\dfrac{E_C - E_{F1}}{0.3} = \dfrac{330k \ln \dfrac{N_C}{N_D}}{300k \ln \dfrac{N_C}{N_D}}$

Hence $(E_C - E_{F1}) = \dfrac{330 \times 0.3}{300} = 0.33\,eV$

Thus, Fermi level lies 0.33 eV below the conduction band.

MASS–ACTION LAWS

- If a pure semiconductor is doped with pentavalent impurities, the number of electron increases but number of holes decreases, as the recombination of electron–hole pair increases
- If a pure semiconductor is doped with trivalent impurities, the number of holes increases but number of electron decreases.
- The law states that : Under thermal equilibrium the product of free electron concentration and hole concentration is constant and is independent of amount of doping. i.e. $n_o p_o = n_i^2$

Mass Action Law

A fundamental relationship between the electron and hole concentrations in a semiconductor in thermal equilibrium is given by

$$n_o p_o = n_i^2 \qquad \text{...(28)}$$

where n_o is the thermal equilibrium concentration of free electrons, p_o is the thermal equilibrium concentration of holes, and n_i is the intrinsic carrier concentration. At room temperature (T = 300 °K), each donor atom donates a free electron to the semiconductor. If the donor concentration N_d is much larger than the intrinsic concentration, we can approximate,

$$n_o \cong N_d$$

Then, from Equation (28), the hole concentration is

$$p_o = \frac{n_i^2}{N_d}$$

Similarly, at room temperature, each acceptor atom accepts a valence electron, creating a hole. If the acceptor concentration N_a is much larger than the intrinsic concentration, we can approximate

$$p_o \cong N_a$$

Then, from Equation (28), the electron concentration is

$$n_o = \frac{n_i^2}{N_a}$$

Example :

Calculate the thermal equilibrium electron and hole concentrations. Consider silicon at T = 300 °K doped with phosphorus at a concentration of $N_d = 10^{16}\ cm^{-3}$ use $n_i = l.5 \times 10^{10}\ cm^{-3}$.

Solution :

Since $N_d >> n_i$, the electron concentration is

$$n_o \cong N_d = 10^{16}\ cm^{-3}$$

and the hole concentration is

$$p_o = \frac{n_i^2}{N_d} = \frac{\left(1.5\times10^{10}\right)^2}{10^{16}} = 2.25\times10^4\ cm^{-3}$$

Note : In an extrinsic semiconductor, the electron and hole concentrations normally differ by many orders of magnitude.

Thus in an n-type semiconductor, electron are called the *majority carrier* because they far outnumber the holes, which are termed the *minority carrier*. In contrast, in a p-type semiconductor, the holes are the majority carrier and the electrons are the minority carrier.

DRIFT AND DIFFUSION CURRENTS

The two basic processes which cause electrons and holes to move in semiconductor are: (a) *drift*, which is the movement caused by electric fields; and (b) *diffusion*, which is the flow caused by variations in the concentration, that is, concentration gradient. Such gradients can be caused by a non-homogeneous doping distribution.

(i) Drift Currents

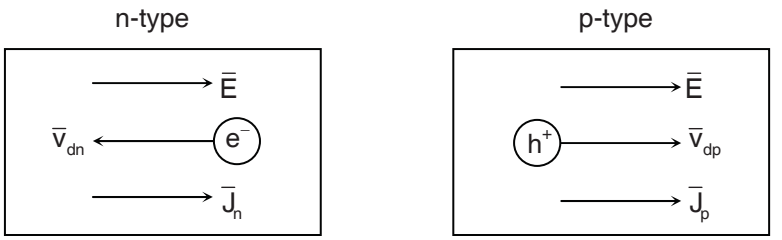
To understand drift, assume an electric field is applied to a semiconductor.

For n–type semiconductor

Consider an n–type semiconductor with a large number of free electrons (following figure (a)). An electric field E applied in one direction produces a force on the electrons in the opposite direction, because of the electron's negative charge. The electrons acquire a net drift velocity v_{dn} (in cm/s) which can be written as

$$v_{dn} = -\mu_n E$$

where μ_n is a constant called the electron mobility and the value of μ_n is typically 1350 cm^2/V–s.



(a) an n-type semiconductor (b) an p-type semiconductor

Applied electric field, carrier drift velocity, and drift current density

The electron drift produces drift current density J_n(A/cm^2) is given by

$$J_n = -env_{dn} = -en\left(-\mu_n E\right) = +en\mu_n E$$

where n is the electron concentration (#/cm^3) and e is the magnitude of the electronic charge. The conventional drift current is in the opposite direction from the flow of negative charge, which means that the drift current in an n-type semiconductor is in the same direction as the applied electric field.

For p–type Semiconductor

Next consider a p–type semiconductor with a large number of holes shown in above figure (a) . An electric field E applied in one direction produces a force on the holes in the same direction because of the positive charge on the holes. The holes acquire a drift velocity v_{dp} (in cm/s) which can be written as

$$v_{dp} = +\mu_p E$$

where μ_p is a constant called the hole mobility and the value of μ_p is typically 480 $cm^2/V–s$.

The hole drift produces a drift current density J_p (A/cm^2) given by

$$J_p = +epv_{dp} = +ep(+\mu_p E) = +ep\mu_p E$$

where p is the hole concentration ($\#/cm^3$) and e is again the magnitude of the electronic charge. The conventional drift current is in the same direction as the flow of positive charge, which means that the drift current in a p-type material is also in the same direction as the applied electric field.

Since a semiconductor contains both electrons and holes, the total drift current density is the sum of the electron and hole components. The total drift current density is then written as

$$\boxed{J = en\mu_n E + ep\mu_p E = \sigma E} \quad \ldots(29)$$

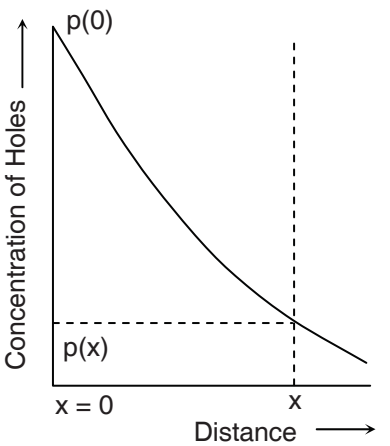
where, $\sigma = en\mu_n + ep\mu_p$ …(30)

and where σ is the *conductivity* of the semiconductor in $(\Omega–cm)^{-1}$. The conductivity is related to the concentration of electrons and holes.

From equation (30), we see that the conductivity can be changed from strongly n-type, n>> p, by donor impurity doping to strongly p-type, p >> n, by acceptor impurity doping. Being able to control the conductivity of a semiconductor by selective doping is what allows us to fabricate the variety of electronic devices that are available.

(ii) Diffusion Current

In a semi-conductor, it is possible to have a non-uniform concentration of particles as shown in figure. Here the concentration p of holes varies with distance x causing a concentration gradient $\frac{dp}{dx}$ in the density of carriers.



Concentration gradient dp/dx resulting diffusion current

Now let us consider surface at distance x. Obviously the density of holes on one side of the surface is larger than the density on the other side. The holes move about randomly because of their thermal energy. The random motion implies that the holes move back forth across the arbitrary surface. Then in a given time interval, more holes cross the surface from the side of greater density to the side of lower density than in the opposite direction. Accordingly net flow of holes takes place across the surface constituting a current in the positive x direction.

This net flow of charge does not result from mutual repulsion among similar charges but is simply a statistical phenomenon. The diffusion hole current density J_D, (amperes/sq. metre) is proportional to the concentration gradient $\frac{dp}{dx}$ and is given by,

$$J_D = -qD_p \frac{dp}{dx} \qquad \ldots(31)$$

where D_P (sq. metres/sec) is called the *diffusion constant* for holes, and q is the magnitude of the charge of hole or electron. It may be noted that the diffusion of carriers results irrespective of the charges carried by the carriers.

In above figure, since hole concentration p decreases with increasing x, $\frac{dp}{dx}$ is negative and a minus sign is, therefore, put in equation (31) to make J_D positive in positive x direction.

Similarly the diffusion current density J_n is given by,

$$J_n = qD_n \frac{dn}{dx}$$

where D_n is the diffusion constant for electrons expressed in metre^2/ second.

Note : The phenomenon of diffusion does not ordinarily take place in metals

- **Total Current in a Semi-conductor**

Within a semi-conductor, both potential gradient (causing drift) and concentration (causing diffusion) may exist simultaneously. In such a case, the total hole current is the sum of the drift current as caused by potential gradient and the diffusion current as caused by hole concentration gradient. Thus total hole current density is

$$J_p = q\,\mu_p\,p\varepsilon - q\,D_p \frac{dp}{dx}$$

Similarly total electron current density is,

$$J_n = q\mu_n n\varepsilon + qD_n \frac{dn}{dx}$$

(iii) The Mobility

The mobility μ of a charge carrier varies as T^{-m} over a temperature range of 100 to 400°K. For silicon, m = 2.5 (2.7) for electron (hole). For germanium, m = 1.66(2.33) for electrons (holes). The mobility μ is also a function of electric field intensity and it remains constant only if $\bar{E} < 10^3$ V/m in n-type silicon. For $10^3 < \bar{E} < 10^4$ V/cm, μ_n in n-type silicon varies approximately as $\bar{E}^{-\frac{1}{2}}$. For higher fields μ_n varies inversely as $\bar{E}$ and the carrier speed approaches a constant value of about 10^7 cm/second.

Einstein Relationship

Both diffusion constant D and mobility μ are statistical thermodynamic phenomena. Hence they are not independent of each other. In fact they are related by the Einstein equation below,

$$\frac{D_D}{\mu_p} = \frac{D_n}{\mu_n} = V_T = \frac{k_T}{q} \quad \ldots \begin{bmatrix} V_T = \text{thermal voltage (26mV @300°K)} \\ T = \text{Temperature in °kelvin} \\ q = \text{electronic charge} \end{bmatrix}$$

Example:

A sample of Germanium is doped to the extent of 10^{14} donor atoms/cm^3 and 5×10^{13} acceptor atoms/cm^3. If the resistivity of intrinsic germanium is 50 Ωcm. If the applied electric field is 2V/cm. Find the total conduction current density. Assume $\frac{\mu_p}{\mu_n} = \frac{1}{2}$ and $n_i = 2.5 \times 10^{13} / cm^3$.

Solution:

$$\sigma = qn_i\left(\mu_n + \mu_p\right)$$

$$\frac{1}{50} = 1.6 \times 10^{-19} \times 2.5 \times 10^{13}\left(3\mu_p\right)$$

$$\mu_p = \frac{1}{50 \times 1.6 \times 10^{-19} \times 2.5 \times 10^{13} \times 3}$$

$$\mu_p = 1667 \text{ cm}^2/\text{Vs}$$

$$\therefore \quad \mu_n = 3334 \text{ cm}^2/\text{Vs}$$

$$p - n = N_A - N_D = 5 \times 10^{13} - 10^{14} = -5 \times 10^{13}$$

$$pn = n_i^2 = \left(2.5 \times 10^{13}\right)^2$$

$$p - \frac{6.25 \times 10^{26}}{p} = -5 \times 10^{13}$$

$$\therefore \quad p^2 + 5 \times 10^{13} p - 6.25 \times 10^{26} = 0$$

On finding the roots, we get

$$p = 1.0356 \times 10^{13}$$

$$\therefore \quad -n = -6.0356 \times 10^{13}$$

$$\therefore \quad n = 6.0356 \times 10^{13}$$

$$\therefore \quad J = q\left(n\mu_n + p\mu_p\right)E$$

$$J = 70 \text{ mA/cm}^2$$

CARRIER GENERATION AND RECOMBINATION

In intrinsic semiconductor, the number of free electrons equals the number of holes since a hole can be created only by removing an electron from a covalent bond. These free electrons and holes move randomly within the crystal lattice. When one such free electron meets a hole, they recombine to re-establish the covalent bond. Both the hole and the free electron then vanish.

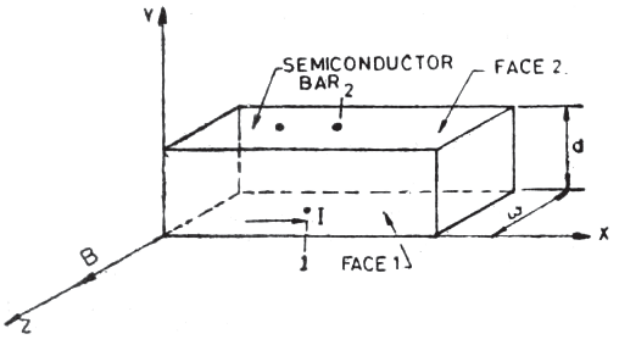
Such a recombination, of course, results in release of energy in the form of heat. The energy so released equals that required for breaking the covalent bond. The rate of recombination is proportional to the product of the concentrations or densities of free electrons and holes. Thus the process of breaking of covalent bonds and recombination of electrons and holes take place simultaneously. Any individual carrier of charge, either electron or hole, then has a limited lifetime typically varying from 1 μ-sec to 1 milli-sec.

At a particular temperature, the densities of electrons and holes remain fixed and equal. Evidently the rate of production of electrons and holes must equal the rate of their recombination. If now the temperature is raised, the rate of production of electrons and holes increases. The densities of electrons and holes then go on increasing until the resulting increased rate of recombination equals the increased rate of production. Thus as the temperature increases, the densities of electrons and holes increase.

HALL EFFECT

When any specimen carrying a current I is placed in a transverse magnetic field B, then an electric field E is induced in the specimen in the direction perpendicular to both I and B. This phenomenon is called the *Hall Effect*. This Hall effect may be used for :

(i) determining whether a semiconductor is p–type or n–type
(ii) finding the carrier concentration and
(iii) calculating the mobility μ, having measured the conductivity σ.



Current carrying semiconductor bar subjected to transverse magnetic field

Above figure shows a semiconductor bar carrying a current $\bar{I}$ in the positive x-direction. Let a magnetic field $\bar{B}$ be applied in the positive z-direction. Then as per Hall effect, a force gets exerted on the charge carriers (whether electrons holes) in the negative y-direction. irrespective of nature of the charge carriers whether holes or electrons, these charge carriers get pressed downwards toward face 1 of the specimen shown in above figure.

In an n-type specimen, electrons accumulate on side 1 which surface then gets negatively charged relative to side 2. A potential difference, therefore, develops between surface 1 and 2 and is called the *Hall voltage*. This Hall voltage V_H in an n-type semiconductor is positive at terminal 2. On the other hand, in a p-type specimen, the Hall voltage V_H is positive at terminal 1.

Experimental Determination of Mobility

In the specimen of above figure, equilibrium condition requires that the electric field intensity E due to the Hall effect exert a force on the carrier equal in magnitude to that exerted by the magnetic field.

Thus,

$$qE = Bqv \qquad \ldots(32)$$

where q is the magnitude of the charge of electron or hole, and v is the drift velocity,

but, $$E = \frac{V_H}{D} \qquad \ldots(33)$$

where d is the distance between the two surfaces 1 and 2.

But current density $\overline{J}$ is given by,

$$\overline{J} = Q_v \; v = \frac{\overline{I}}{wd} \qquad \ldots(34)$$

where Q_v is the charge density and w is the width of the specimen along the magnetic field.

Combining equations (32), (33) and (34), we get

$$V_H = Ed = Bvd = \frac{BJd}{Q_v} = \frac{BI}{Q_v w} \qquad \ldots(35)$$

Now quantities V_H, B, I and w for a specimen may be measured and hence from equation (35), Q_v can be determined. Knowing Q_v, the charge density (Q_v/q) may be computed.

The Hall coefficient R_H is defined as

$$R_H \equiv \frac{1}{Q_v}$$

Hence equation (35) yields,

$$R_H = \frac{V_H w}{BI}$$

In any extrinsic semiconductor, the conductivity is due primarily to charge of only one sign. In that case, conductivity σ is given by,

$$\sigma = Q_v \mu$$

If both σ and R_H are measured, the mobility may be determined from the following relation :

$$\mu = \sigma R_H$$

Application of Hall Effect

(i) *Magnetic field meter*
The Hall voltage V_H for a given current, is proportional to B. Hence measurement of V_H measures the magnetic field B.

(ii) *Hall effect multiplier*
This instrument gives an output proportional to the product of two signals. Thus if current I is made proportional to one input and if B proportional to the second input, then as per equation (35), Hall voltage V_H is proportional to the product of the two inputs.

Example:

Consider a semiconductor with dimensions (0.9 × 0.4 × 0.3) cm and the area of the surface is 0.36 cm^2. If the specimen is applied a magnetic field of 6×10^{-5} Wb/m^2 and the drift velocity of the carriers is 10^7 m/s, then calculate the hall voltage

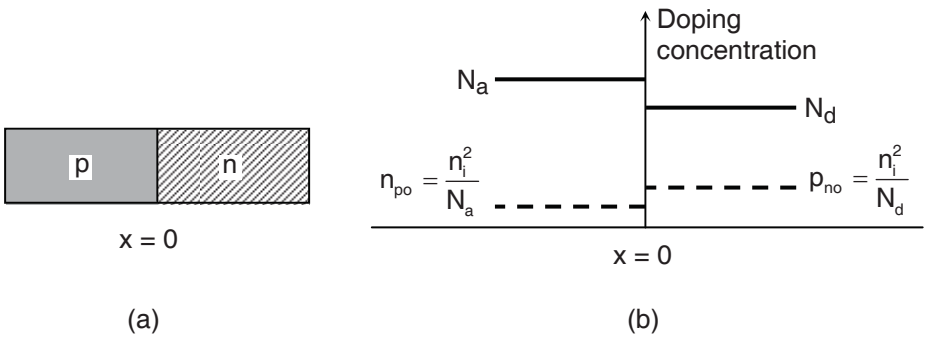
Solution:

We know, Hall voltage is given by

$$V_H = Bvd$$
$$= 6 \times 10^{-5} \times 10^7 \times 0.3 \times 10^{-2}$$
$$V_H = 1.8 \text{ V}$$

P–N JUNCTION AS A DIODE

The real power of semiconductor electronics occurs when p- and n-regions are directly adjacent to each other, forming a *pn junction*.

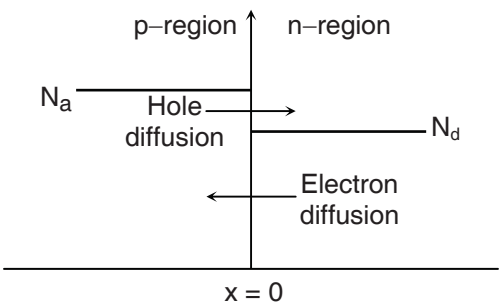


The pn junction : (a) simplified geometry of a pn junction and (b) doping profile of an ideal uniformly doped pn junction

(i) The Equilibrium pn Junction (No External Battery Voltage)

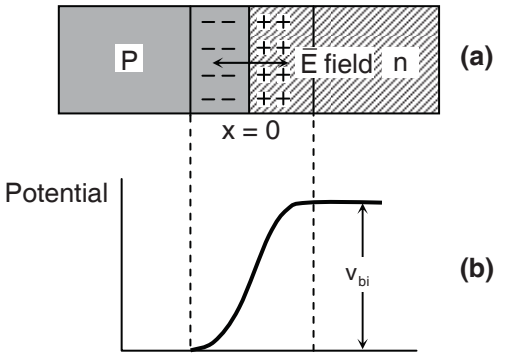
Above figure (a) is a simplified block diagram of a pn junction. Above figure (b) shows the respective p-type and n-type doping concentrations assuming thermal equilibrium.

The interface at x = 0 is called the *metallurgical junction*. A large density gradient in both the hole and electron concentrations occurs across this junction, because of this there is a diffusion of holes from the p-region into the n–region, and a diffusion of electrons from the n-region into the p-region (refer adjoining figure). The flow of holes from the p–region uncovers negatively charged acceptor ions, and the flow of electrons from the n-region uncovers positively charged donor ions.



Initial diffusion of electrons and holes at the metallurgical junction, establishing thermal equilibrium

This action creates a charge separation (following figure (a)), which sets up an electric field oriented in the direction from the positive charge to the negative charge. If no voltage is applied to the pn junction, the diffusion of holes and electrons must eventually cease. The direction of the induced, electric field will cause the resulting force to repel the diffusion of holes from the p-region and the diffusion of electrons from the n-region. Thermal, equilibrium occurs when the force produced by the electric field and the "force" produced by the density gradient exactly balance.



The pn junction in thermal equilibrium (a) the space-charge region and electric field and (b) the potential through the junction

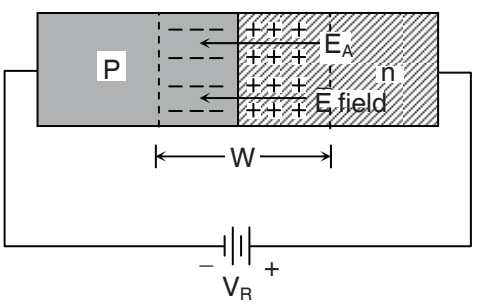
The positively charged region and the negatively charged region comprise the *space-charge* region, or *depletion region*, of the pn junction, in which there are essentially no mobile electron or holes. Because of the electric field in the space-charge region, there is potential difference across that region (refer above figure (b)). This potential difference is called the *built-in potential barrier*, or built–in vo1tage, and is given by

$$V_{bi} = \frac{kT}{q} \ln\left(\frac{N_a N_d}{n_i^2}\right) = V_T \ln\left(\frac{N_a N_d}{n_i^2}\right) \qquad \ldots(36)$$

where $V_T \equiv kT/q$, k = Boltzmann's constant, T = absolute temperature, q = the magnitude of the electronic charge, and N_a and N_d are the net acceptor and donor concentrations in the p- and n-regions, respectively. At room temperature V_{bi} is usually in the range of 0.6 to 0.7 V for silicon.

(ii) Reverse-Biased pn Junction

Assume a positive voltage is applied to the n-region of a pn junction, as shown in adjoining figure. The applied voltage V_R induces an applied electric field, E_A in the semiconductor. The direction of this applied field is the same as that of the E-field in the space-charge region. The magnitude of the electric field in the space-charge region increases above the thermal equilibrium value. This increased electric field holds back the holes in the p-region and the electrons in the n-region, so there is essentially no current across the pn junction. By definition, this applied voltage polarity is called *reverse bias*.



A pn junction with an applied revesre–bias voltage, showing the direction of the electric field induced by V_R and of the space–charge electric field

If the doping concentrations are not changed, the increases in the charges can only occur if the width W of the space-charge region increases. Therefore, with an increasing reverse-bias voltage V_R, space-charge width W also increases.

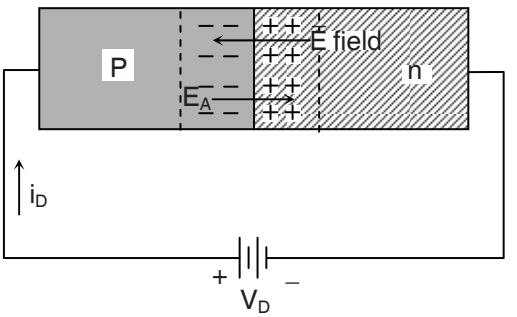
Because of the additional positive and negative charges in the space–charge region, a capacitance is associated with the pn junction when a reverse–bias voltage is applied. This *junction capacitances*, or depletion layer capacitance, can be written in the form

$$C_j = C_{jo}\left(1+\frac{V_R}{V_{bi}}\right)^{-1/2}$$

where C_{jo} is the junction capacitance at zero applied voltage.

(iii) Forward–Biased pn Junction :

If a positive voltage v_D is applied to the p–region, the potential barrier decreases (refer following figure). This applied voltage exists across the pn junction region. The applied electric field, E_A, induced by the applied voltage is in the opposite direction from that of the thermal equilibrium space–charge E–field. The net result is that the electric field in the space–charge region is lower than the equilibrium value. This upsets the delicate balance between diffusion and the E-field force. Majority carrier electrons from the n–region diffuse into the p-region, and majority carrier holes from the p–region diffuse into the n-region. The process continues as long as the voltage v_D is applied, thus creating a current in the pn junction. This applied voltage polarity (i.e., bias) is known as *forward bias*.



A pn junction with an applied forward–bias voltage, showing the direction of the electric field E_A induced by v_D and of the net space–charge electric field E.

Note : The forward–bias voltage v_D must always be less than the built-in potential barrier V_{bi}.

The current voltage relationship of pn-junction diode is given by

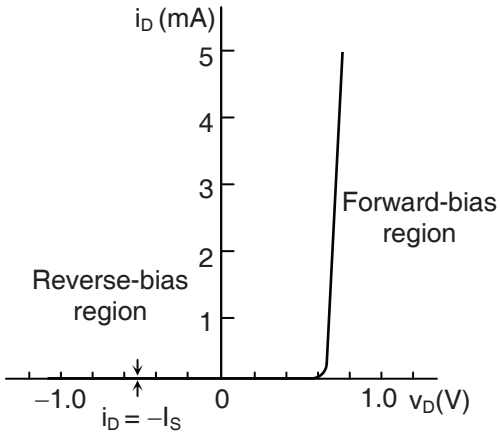
$$i_D = I_S\left[e^{\left(\frac{v_D}{nV_T}\right)}-1\right] \qquad \ldots(37)$$

The parameter I_S is the *reverse-bias saturation current*. For silicon pn junctions, typical values of I_S are in the range of 10^{-15} to 10^{-13} A. The parameter V_T is the thermal voltage which is equal to 0.026V at room temperature. The parameter n is usually called the emission coefficient or ideality factor, and its value is in the range $1 \le n \le 2$.

The emission coefficient n takes into account any recombination of electrons and holes in the space-charge region. At very low current levels, recombination may be a significant factor and the value of n may be close to 2. At higher current levels, recombination is less a factor, and the value of n will be 1. Unless otherwise stated, we will assume the emission coefficient is n = 1.

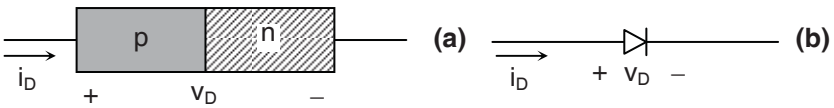
VI CHARACTERISTICS OF DIODE

Following figure is a plot of the derived current–voltage characteristics of a pn junction. For a forward-bias voltage, the current is an exponential function of voltage.



Ideal I–V characteristics of a pn junction diode for $I_S = 10^{-14}$A

The semiconductor device that displays these I–V characteristics is called a *pn junction diode*. Following figure shows the diode circuit symbol and the conventional current direction and voltage polarity. The diode can be thought of and used as a voltage controlled switch that is "off" for a reverse-bias voltage and "on" for a forward–bias voltage.



The basic pn junction diode: (a) simplified geometry and
(b) circuit symbol, and conventional current direction and voltage polarity

Example :
For the silicon diode at 290°K, with current of 0.5mA at voltage 340 mV, determine the reverse saturation current.
Solution:

at 290°K

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times 290}{1.6 \times 10^{-19}} = 24.9\,\text{mV}$$

and assuming $\eta = 1$

$\therefore \quad I_D = I_S\left[e^{\frac{V_D}{\eta V_T}} - 1\right]$

$\therefore \quad 0.5\times10^{-3} = I_S\left[e^{\frac{340}{24.9}}\right]$ …[as $V_D \gg V_T$ we neglect –1]

$\therefore \quad 0.5\times10^{-3} = I_S\left[e^{13.605}\right]$

$\therefore \quad I_S = 0.617\ \mu A$

JUNCTION TEMPERATURE EFFECT

Since both I_S and V_T are functions of temperature, the diode characteristic also vary with temperature. V_T has a negative temperature coefficient and it reduces approximately –1.8 mV/°C for silicon junction and –2.02 mV/°C for germanium.

Reverse saturation current I_S also doubles in magnitude for every 10°C rise in temperature. Value of I_S for silicon are much lower than that of germanium for same levels. As an example of the importance of this effect, in germanium, the relative value of n_i is large, resulting in a large reverse-saturation current in germanium-based diodes. Increases in this reverse current with increases in the temperature make the germanium diode highly impractical for most circuit applications.

As the temperature increases the forward characteristics actually become ideal but beyond the normal operating range it can have harmful effects on the diodes maximum power and current levels. In reverse bias region the increasing value of I_S with temperature account for lower levels of threshold voltage.

BREAKDOWN VOLTAGE

The magnitude of the electric field in the space-charge region increases as the reverse-bias voltage increases, and the maximum electric field occurs at the metallurgical-junction. However, neither the electric field in the space-charge region nor the applied reverse-bias voltage can increase indefinitely because at some point breakdown will occur and a large reverse bias current will be generated.

Zener Breakdown

When a reverse-bias voltage is applied to a pn junction, the electric field in the space-charge region increases. The electric field may become large enough that covalent bonds are broken and electron-hole pairs are created. Electrons are swept to the n-region and holes to the p-region by the electric field generating a reverse-bias current. This breakdown mechanism is called the *Zener effect***.**

Avalanche Breakdown

Another breakdown mechanism is called *avalanche breakdown*, which occurs when minority carriers crossing the space-charge region gain sufficient kinetic energy to be able to break covalent bonds during a collision process. The generated electron-hole pairs can themselves be involved in a collision process generating additional electron-hole pairs, thus, the avalanche process. The reverse-bias current for each breakdown mechanism will be limited by the external circuit.

A pn junction is usually rated in terms of its *peak inverse voltage* or *PIV*. The PIV of a diode must never be exceeded in circuit operation if reverse breakdown is to be avoided. Zener diodes are fabricated with a specifically designed breakdown voltage and are designed to operate in the breakdown region. These diodes are discussed later in this chapter.

Silicon Vs Germanium

Silicon has higher PIV and current rating, wider temperature ranges than germanium diodes.

	Silicon	Germanium
• PIV ratings	1000 V	400 V
• Temperature	200 °C (400 °F)	100 °C
• Threshold Voltage	0.7 V	0.3 V

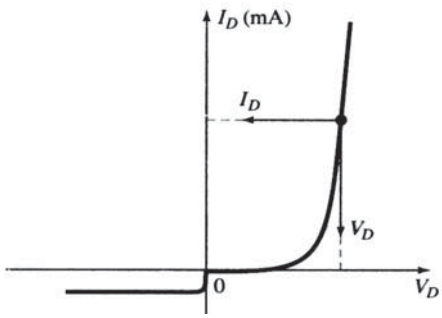
- Breakdown voltage increases with temperature but result in undesirable increase in reverse saturation current.
- Drawback of silicon is its higher forward bias voltage required to reach the region of upward swing.
- If greater PIV rating is required, a number of diodes of same characteristics should be connected in series.

RESISTANCE LEVELS

The resistance of the diode changes as the operating point of a diode moves from one region to another

- **DC or static resistance**:
  Application of dc voltage to a semiconductor diode circuit will result in operating point that will not change with time.

$$r_f = \frac{V_D}{I_D}$$



Determining the dc resistance of a diode at a particular operating point

Example:

Determine the dc resistance levels for the diode

Solution:

(a) For forward biased

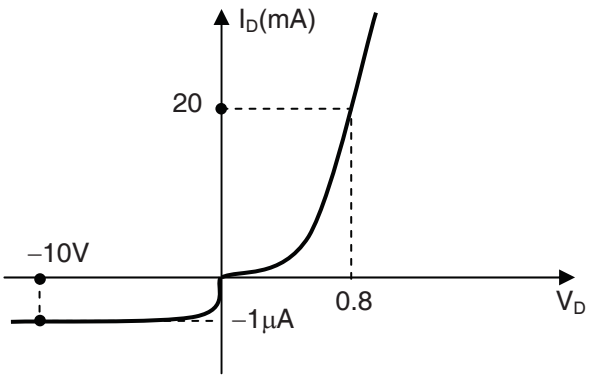
$I_D = 20$ mA, $V_D = 0.8$V

$$\therefore \quad r_f = \frac{V_D}{I_D} = \frac{0.8V}{20mA} = 40\Omega$$

(b) For reverse biased

$I_D = -1\mu A$, $V_D = -10V$

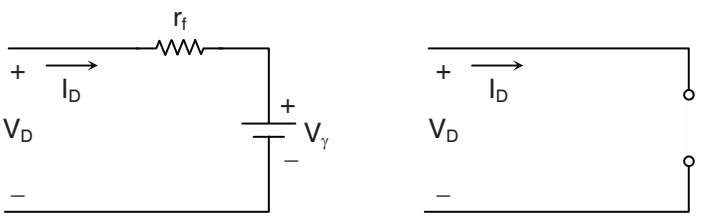
$$\therefore \quad r_f = \frac{V_D}{I_D} = \frac{-10V}{-1\mu} = 10M\Omega$$



Thus from above example, we can model the diode as

- **Piecewise Linear Model**

For $V_D \geq V_\gamma$, we assume a straight-line approximation whose slope is $1/r_f$, where V_γ is the *turn-on*, or *cut-in*, *voltage* of the diode, and r_f is the *forward diode resistance*. The equivalent circuit for this linear approximation is a constant-voltage source in series with a resistor (following figure (a)). For $V_D < V_\gamma$, we assume a straight-line approximation parallel to the V_D axis at the zero current level. In this case, the equivalent circuit is an open circuit (following figure (b)).

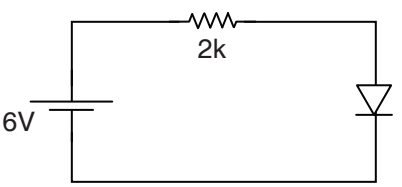


(a) (b)

The diode equivalent circuit (a) in the "on" condition when $V_D \geq V_\gamma$,
(b) in the "off" condition when $V_D < V_\gamma$

Example:

In the circuit shown the diode has a threshold voltage of 0.5V and a forward resistance of 1Ω. Find the operating point of the diode.



Solution :
Replacing diode by its model, we get

$$-6+2\times10^3 I_D+1\times I_D+0.5=0$$

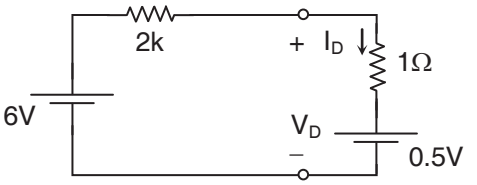
$$\therefore \quad I_D(2001)=6-0.5$$

$$\therefore \quad I_D=\frac{6-0.5}{2001}=2.75\text{mA}$$

$$V_D=R_F I_D+V_F$$

$$=1\times2.75\times10^{-3}+0.5$$

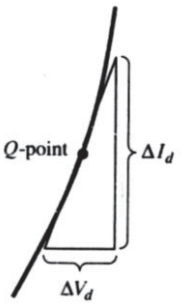
$$V_D=0.5028\text{V}$$



- **AC or Dynamic resistance**
  Application of Sinusoidal input will move the instantaneous operating point up and down on the characteristic curve. With no applied varying signal, the point of operation would be Q–point i.e. quiescent point

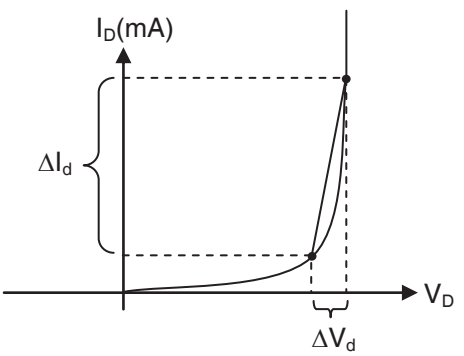
$$r_d=\frac{\Delta V_d}{\Delta I_d}$$

Δ : signifies a finite change in the quantity



Determining the ac resistance at a Q–point

- **Average AC Resistance**
  If the input signal is sufficiently large to produce a broad swing, then the resistance associated with the device for this region is called the average ac resistance.



CAPACITANCE IN P–N JUNCTION

The biasing and unbiasing operation results in charge accumulation across the depletion region. This charge accumulation results in the capacitance action. The depletion area acts as the dielectric media between parallel plates of the capacitor.

There are two main types of capacitance in pn junction diodes.
(A) Depletion (Transition) capacitance C_T
(B) Diffusion capacitance C_D

(A) Depletion Capacitance C_T

It is due to space charge in the depletion layer. The depletion width is a function of a bias voltage i.e. as reverse voltage increases, the depletion width also increases

It is given by

$$C_T = \frac{\in A}{W} \text{ (F)}$$

A : Cross sectional area. (m^2)

W : Thickness (m)

$\in$: Permittivity (F/m)

Alternatively : Refer section on reversed biased PN junction.

Example:

In case of the diode, the depletion layer is 1.2×10^{-6} cm, the area of cross section is 2 mm^2 and the relative permittivity of the medium is 13, then what will be the diffusion capacitance

Solution:

We know $$C_T = \frac{\in A}{d} = \frac{13 \times 8.854 \times 10^{-14} \times 2 \times 10^{-2}}{1.2 \times 10^{-6}} = 191.83 \times 10^{-10}$$

$$C_T = 19.2 \times 10^{-9} = 19.2 \text{ nF}$$

(B) Diffusion capacitance, C_D

It is due to accumulation of free charges on either side of the junction under forward bias condition. Transition capacitance has no meaning if the junction potential vanishes. The majority carriers starts moving to other region i.e. holes towards n–type and electrons towards p–type. This results in accumulation of charges on either side resulting in diffusion capacitance C_D.

There is a time lag between change of bias. This time is necessary to charge or discharge diffusion capacitance.

It is given by $$C_D = \frac{\tau I}{\eta V_T}$$

Example:

In case of germanium diode for the current of 26mA, τ = 20 μsec., then what will be diffusion capacitance.

Solution :

We know

$$C_D = \frac{\tau I}{\eta V_T}$$

Here τ = 20 μsec, I = 26 mA, η = 1

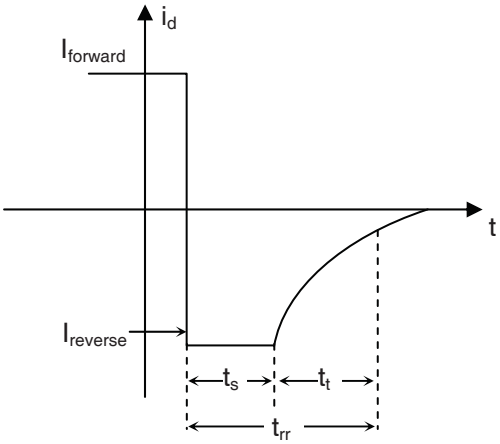
V_T = 0.3V

$$\therefore \quad C_D = \frac{20 \times 10^{-6} \times 26 \times 10^{-3}}{1 \times 0.3} = 1.74 \times 10^{-6}$$

C_D = 1.74 μF

Note : Generally $C_D >> C_T$, as shown from above example.

DIODE SWITCHING TIMES



- If a forward voltage is applied to the diode and if this voltage is reversed to establish reverse bias condition then ideally the diode should change instantaneously from conducting to non conducting state. However due to large number of minority carriers in each material, the diode simply reverses and remains in that state till the minority carriers return to majority carrier state in opposite material. This period is the storage time t_s.
- After the storage phase has passed, the current reduces to the level associated with non–conduction state. This period is the transition time t_t.
- Reverse recovery time is the sum of the two intervals

  $t_{rr} = t_s + t_t$

BREAKDOWN DIODES OR ZENER DIODES

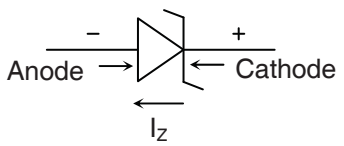
There are two mechanisms that causes breakdown in a reverse biased pn junction. With a very narrow depletion region, the electric field strength (V/width) produced by a reverse voltage can be very high. The high intensity electric field causes electrons to break away from their atoms, thus converting the depletion region from an insulating material into a conductor. This is ionization by electric field also called *zener breakdown.* Where the depletion region is too wide for zener breakdown, the electrons reverse saturation current can be given sufficient energy to cause other electrons to break free when they strike atoms within the depletion region. The electrons released in this way collide with other atoms to produce more free electrons in an avalanche effect. *Avalanche breakdown* is normally produced by reverse voltage above 5V.

Circuit Symbol of Zener Diode

In breakdown region, large changes in diode current produce small changes in diode voltage. Upper limit on diode current is determined by the power dissipation rating.

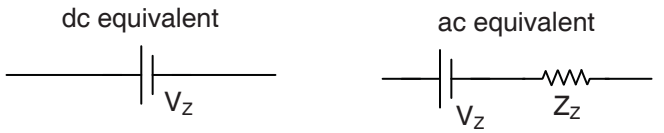
Dynamic Resistance Z_Z ,

$$Z_Z = \frac{\Delta V_Z}{\Delta I_Z}$$



Equivalent Circuit

Note that these equivalent circuits are applied only when the Zener diode is maintained in reverse breakdown. If the device becomes forward biased, then the equivalent circuit for a forward biased diode must be used. (normal PN junction diode)



Zener is commonly used in applications involving fixed voltage or reference voltage requirements.

- By the application of thermal energy, new carriers are produced. These new carriers collide with the other bonded atoms and results into disrupting bonds. More electron–hole pairs are generated. Thus cumulative process is called as Avalanche multiplication.

  Thus large reverse current flows and diode enters into avalanche breakdown.

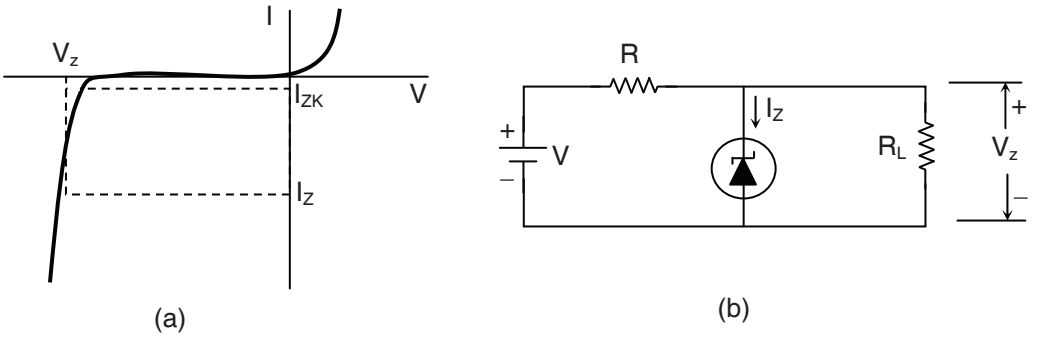


Fig (a) The volt ampere characteristic of an avalanche, or Zener diode,
(b) A circuit in which such a diode is used to regulate the voltage across R_L against change in the variation in load current and supply voltage.

- Strong electric field also disrupts the bond under these circumstances the breakdown is referred to as Zener breakdown

> Avalanche voltage increases with increased temperature.

In case of broad depletion layer, we rely on intrinsic carriers to collide with valence electrons to create avalanche. As temperature increases the vibration of electron increases and collision probability increases. The intrinsic holes and electrons thus have less opportunity to gain sufficient energy before collision to activate avalanche. Hence Avalanche voltage increases with increase in temperature.

Example :
Consider a simple constant-voltage reference circuit and design the value of resistance required to limit the current in this circuit. Consider the circuit shown in following figure. Assume that the Zener diode breakdown voltage is V_Z = 5.6 V and the Zener resistance is r_z = 0. The current in the diode is to be limited to 3 mA.

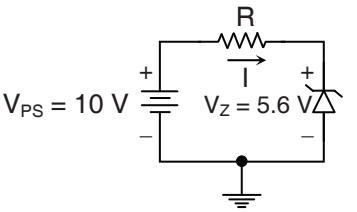
Solution :
As before, we can determine the current from the voltage difference across R divided by the resistance.

That is,

$$I = \frac{V_{PS} - V_Z}{R}$$

The resistance is then

$$R = \frac{V_{PS} - V_Z}{I} = \frac{10 - 5.6}{3} = 1.47\,k\Omega$$



Simple circuit containing a Zener diode

LIST OF FORMULAE

- Hall effect : $V_H = Bvd$
- Current density J $= (n\mu_n + p\mu_p)eE$
  $= \sigma E$
  where,
  n = Magnitude of free electron concentration (cm^{-3})
  σ = Conductivity ($\mho / Cm$)
  p = Magnitude of hole concentration. (cm^{-3})
- Intrinsic concentration n_i = n=p for pure semiconductor OR intrinsic semiconductor.

$$n_i = BT^{3/2} e^{\left(\frac{-E_G}{2kT}\right)}$$

- Mass action law : $n_o p_o = n_i^2$
- Fermi level :

(a) $E_F = \frac{E_C + E_V}{V}$ → For intrinsic semiconductor

(b) $E_F = E_C - KT \ln \frac{N_C}{N_D}$ → For n type material

E_C = Conduction band

E_F = Fermi level

E_V = Valence band

(c) $E_F = E_V + KT \ln \frac{N_V}{N_A}$ → For p type material

- $N_C = 2\left(\frac{2\pi m_n KT}{h^2}\right)^{3/2}\left(1.6x10^{-19}\right)^{3/2}$
- $N_v = 2\left(\frac{2\pi m_p KT}{h^2}\right)^{3/2}\left(1.6x10^{-19}\right)^{3/2}$ …[K = Boltzman constant in ev/°K]

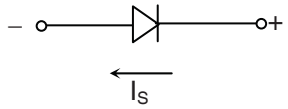
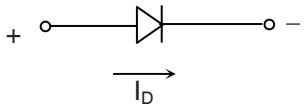
  $m_n, m_p \rightarrow$ effective mass of electron , holes respectively
- $C_T = \frac{\in A}{w}$
- $C_D = \frac{\tau I}{\eta V_T}$
- $V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$
- Einstein relation : $\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = V_T = \frac{kT}{q}$ …[K = Boltzman constant in joule/°K]

IMPORTANT TERMS

- Conductor : Any material that will support a large flow of charge after application of a potential difference of limited magnitude.
- Insulator : Any material that offers a very low level of conductivity.
- Semiconductor : Conductivity between the extremes of an insulator and a conductor.
- Covalent bond : Bonding of atoms strengthened by the sharing of electrons.
- Intrinsic : Semiconductors that have been carefully refined to reduce impurities i.e. as pure as Material can be made available
- Extrinsic : Semiconductor material that has been subjected to doping process Material
- Donor atoms : Diffused impurities with five valence electrons.
- Acceptor atoms : Diffused impurities with three valence electrons.
- PIV : The maximum reverse bias potential that can be applied before entering the Zener region is called peak inverse voltage (PIV) or peak reverse voltage (PRV).

LMR (LAST MINUTE REVISION)

- Germanium and silicon have gap energies E_g of 0.785 and 1.21 eV at $0°k$ and 0.07 and 1.l eV at room temperature 300° k
- Electron energy level is referred in terms of electron volt ($leV = 1.6x10^{-19} J$)
- Biasing Symbol

No	$V_D = 0V$
Reverse	V_D 
Forward	V_D 

- Fermi energy level changes as doping concentration changes.
- Mass action law $np = n_i^2$
  Product of free electron and hole concentration is constant under thermal equilibrium.
- In n–type semiconductor electrons are majority carriers. In p–type semiconductor holes are majority carriers.

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ASSIGNMENT – 1

Duration : 45 mins **Maximum Marks : 30**

Q1 to Q6 carry one mark each

1. If the current of a Si diode with V_T = 26mV is increased four times then the increase in forward voltage drop is _____

(A) 36.04 mV (B) –36.04 mV
(C) 72.087 mV (D) –72.087 mV

2. Transition capacitance of a varactor diode is C_T. The width of depletion layer is 4×10^{-6} cm and cross sectional area is 1mm^2 with relative permittivity of medium as 12 then transition capacitance is given by _____

(A) 2.62×10^{-10} F (B) 2.656 nF
(C) 2.26 nF (D) none of these

3. If biasing of diode is changed instantaneously from forward to reverse the current does not flow immediately but takes some time to change. This time is called as_____

(A) delay time (B) storage time
(C) reverse recovery time (D) (A) and (C) both

4. A n type silicon sample of 5 mm long has rectangular cross section 50 × 100 μm. The donor concentration at 300°K is 5×10^{14} cm^{-3} and corresponds to 1 impurity atom for 10^8 silicon atoms. A steady current of 1 μA exists in the bar. Determine the electron and hole concentration. [$N_i = 1.45 \times 10^{10}$, $\mu_n = 1.5 \times 10^3$]

(A) 5×10^{14} cm^{-3}, $4.2 \times 10^{+5}$ cm^{-3} (B) 5×10^{14} cm^{-3}, $2.9 \times 10^{+5}$ cm^{-3}
(C) $2.9\times10^{+5}$ cm^{-3}, 5×10^{14} cm^{-3} (D) None of these

5. The depletion capacitance C_p of an abrupt P–N junction with constant doping on either side varies with reverse bias, V_R as

(A) $C_J \propto V_R$ (B) $C_J \propto V_R^{-1}$
(C) $C_J \propto V_R^{-1/2}$ (D) $C_J \propto V_R^{-1/3}$

6. The time for which hole or electron exists before recombination is called as _____

(A) minimum lifetime (B) mean lifetime
(C) shell lifetime (D) maximum lifetime

Q7 to Q18 carry two marks each

7. Measurement of Hall coefficient enables the determination of

(A) recovery time of stored carrier
(B) type of conductivity and concentration of charge carriers
(C) temperature coefficient and thermal conductivity
(D) fermi level and forbidden energy gap

8. The mean free path of conduction electrons in copper is about 4×10^{-8} m. For a copper block find the electric field which can give 1eV energy to a conduction electron.

(A) $E = 2.62 \times 10^7$ V/m (B) $E = 2.5 \times 10^7$ V/m

(C) $E = 2.64 \times 10^7$ V/m (D) $E = 2.58 \times 10^7$ V/m

9. A Ge diode is operated at a junction temperature of 27°C. For a forward current of 10 mA, V_D is found to be 0.3 V. If V_D = 0.4 V, find the forward current.

(A) 477.3 mA (B) 4.77 mA

(C) 47.7 mA (D) None of these

10. Which of the following equation holds valid for the relation between the density of hole–electron pairs (n_i) and temperature (T) and energy gap E_{G0}?

(A) $n_i = A_0 T^3 e^{-E_{G0}/kT}$ (B) $n_i^2 = A_0 T^3 e^{-E_{G0}/kT}$

(C) $n_i^2 = A_0 T^2 e^{-E_{G0}/kT}$ (D) None of these

11. An intrinsic silicon bar is 3 mm long and has a rectangular cross section 50 × 100 μm. At 300 K, determine the electric field intensity in the bar for current of 1μA.

$\rho = 2.3 \times 10^5\ \Omega.cm$

(A) 4.6×10^5 V/cm (B) 4.6×10^5 V/m

(C) 4.6×10^4 V/cm (D) None of these

12. In a p–n junction diode, if the average time considering both electron flow in conduction band and valence band is doubled, but the forward diode current is made half of its original value then change in storage capacitance will _____

(A) remain same (B) Doubles

(C) reduce by half (D) reduce to one fourth

13. As the doping concentration varies the Fermi level_____

(A) increases (B) decreases

(C) remains same (D) changes

14. In a semiconductor, if the applied electric field is increased, then

Statement 1: mobility of charge carriers increases

Statement 2: drift velocity increases

Choose the correct option

(A) 1–True, 2–True (B) 1–True, 2–False

(C) 1–False, 2–True (D) 1–False, 2–False

15. Two statements are given below

Statement 1 : Depletion layer capacitance is due to accumulation of free charges on either side of junction under forward bias condition.

Statement 2 : Diffusion capacitance is due to the accumulation of space charge in depletion layer.

Choose the correct option

(A) 1–True, 2–True (B) 1–False, 2–True

(C) 1–False, 2–False (D) 1–True, 2–False

16. If the probability of electron being thermally promoted to conduction band is very less of the order of 3.4×10^{-10} ,then the difference between conduction band level and Fermi level at room temp (25 °C) will be,

(A) 6.8×10^{-20} J (B) 0.536 eV
(C) 8.6×10^{-20} J (D) none of these

17. In Hall probe device, the voltage is changed from 4V to $\frac{V}{4}$, then the magnetic field intensity will change from 2B to _____.

(A) B/2 (B) B/4
(C) B/8 (D) B/16

18. Electron mobility and life time in a semiconductor at room temperature are respectively 0.36 m^2/Vsec and 340 μsec. The diffusion length is _____

(A) 3.13 mm (B) 1.78 mm
(C) 3.55 mm (D) 7.18 mm

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TEST PAPER – 1

Duration : 30 mins **Maximum Marks : 25**

Q1 to Q5 carry one mark each

1. If a sample of germanium and a sample of silicon have the same impurity density and are kept at room temperature
(A) both will have equal value of resistivity
(B) both will have equal negative resistivity
(C) resistivity of germanium will be higher than that of silicon
(D) resistivity of silicon will be higher than that of germanium

2. The Hall constant in p–Si bar is given by 5×10^3 cm^3/ coulomb. The hole concentration in the bar is given by :
(A) 1.00×10^{15} / cm^3 (B) 1.25×10^{15} / cm^3
(C) 1.50×10^{15} / cm^3 (D) 1.6×10^{15} / cm^3

3. A heavily doped semiconductor exhibits a _____ such a device is called sensistors.
(A) Negative temperature coefficient of resistance
(B) Positive temperature coefficient of resistance till a particular value of temperature and then it shows negative temperature coefficient of resistance.
(C) Positive temperature coefficient of resistance
(D) None of the above

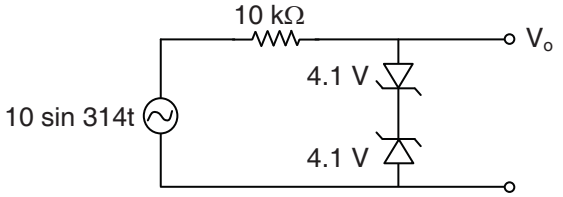
4. For a zener diode, change in temperature is 50°C and temperature co–efficient is 0.072. If the zener potential is 8V then for corresponding change is temperature, what will be the change reflected in zener potential?
(A) 7.98 (B) 0.1 V
(C) 0.8V (D) 0.288 V

5. Suppose an intrinsic semiconductor has 1 billion free electrons at room temperature. If the temperature changes to 75°C, how many holes are there?
(A) fewer than 1 billion (B) 1 billion
(C) more than 1 billion (D) impossible to say

Q6 to Q13 carry two marks each

6. The specific gravity of tungsten is 13.8 and its atomic weight is 184.0. Assume that there are two free electrons per atom. Then the Fermi level or characteristic energy for the crystal in eV will be
(A) 8.95 eV (B) – 8.95 eV
(C) 7.326 eV (D) –7.326 eV

7. The waveshape of V_o in figure will be

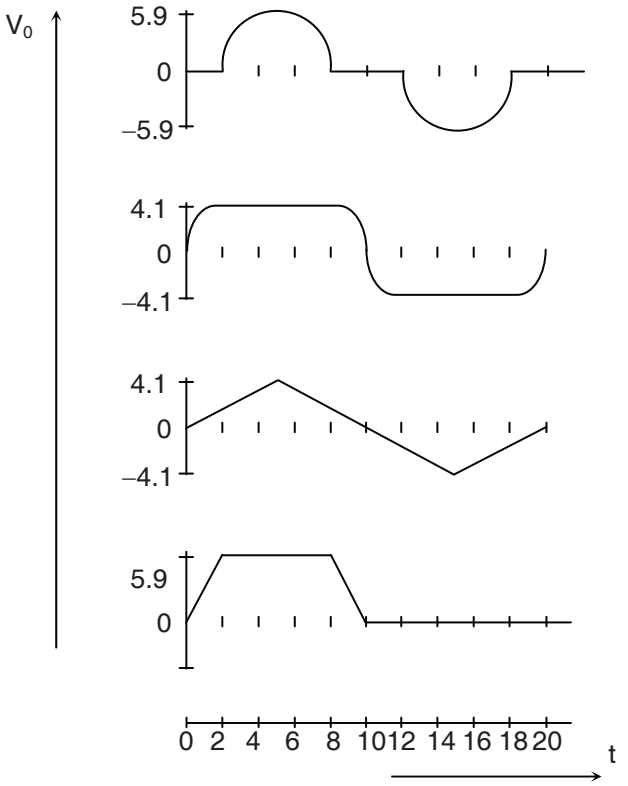


(A)

(B)

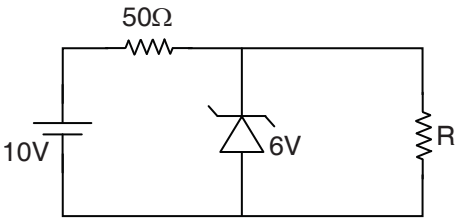
(C)

(D)



8. The 6V Zener diode shown in Figure has zero Zener resistance and a knee current of 5mA. The minimum value of R, so that the voltage across it does not fall below 6V is

(A) 1.2 kΩ
(B) 80 Ω
(C) 50 Ω
(D) 0



9. The intrinsic carrier concentration for a Si specimen at 270 K is _____
Given: B = 5.23×10^{15}

(A) 94.46×10^{8} cm^{-3}
(B) 9.446×10^{9} cm^{-3}
(C) 9.446×10^{6} cm^{-3}
(D) None of these

10. Determine the electron (minority carrier) and hole (majority carrier) concentration in p–type Si with an acceptor ion concentration of 1×10^{16} cm^{-3} when operating at 270°K.
Given: $n_i = 9.446 \times 10^8$ cm^{-3}

(A) 8.92 cm^{-3}, 10^{16} cm^{-3}
(B) 89.2 cm^{-3}, 10^{16} cm^{-3}
(C) 892.0 cm^{-3}, 10^{16} cm^{-3}
(D) None of these

11. For a silicon diode, if energy gap is maintained constant then to double the intrinsic carrier concentration, the temperature has to change by factor _____ from room temperature.

(A) 1.75
(B) 1.875
(C) 1.587
(D) 2

12. Note the following statements : If the donor atom concentration N_D added to a p–type semiconductor that exceeds the acceptor ion concentration then
Statement 1: p–type semiconductor will change to n–type semiconductor.
Statement 2: No change in semiconductor results.
Statement 3: n–type semiconductor will change to p–type semiconductor.
Choose the correct option:

(A) 1 – True, 2 – True, 3 – False
(B) 1 – True, 2 – False, 3 – False
(C) 1 – False, 2 – True, 3 – True
(D) 1 – False, 2 – False, 3 – True.

13. In case of metal A, the average speed of electrons under motion is given by $\frac{V}{2}$ m/s. The electric field which caused the motion is E V/m. Now if the electric field is doubled and applied to metal B with same mobility as the metal A, the average drift speed will change to _____.(Consider mobility of electrons will change negligibly)

(A) V/2
(B) V
(C) 2V
(D) none of these

Q14(a) & (b) carry two marks each

Linked Answer Question

An n–type silicon bar has a majority carrier concentration of 5×10^{20} /m^3 and the carrier mobility is 0.13 m^2/V–s at 300 K.

14(a). If the charge of an electron is 1.6×10^{-19} coulomb, then the resistivity of the bar is

(A) 0.96 Ωm
(B) 0.096 Ωm
(C) 96 Ωcm
(D) 9.6 Ωm

14(b). If the bar is 0.1 cm long and 100 μm^2 in cross–section its resistance will be

(A) 10^6 ohm
(B) 10^4 ohm
(C) 10^3 ohm
(D) 10^{-4} ohm

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